

Using Shiny Numerati Dashboard to Find Your Best Models

Jo-fai (Joe) Chow

Numerai Toronto Community Meetup
2023-10-14



About Jo-fai (Joe) Chow

Background

Civil Engineer (Water Infrastructure)
Taught myself R & Python back in 2012

Work

Data Scientist / Meetup Organizer
(ex- Domino Data Lab & H2O.ai)

Contact

Twitter: @matlabulous
www.linkedin.com/in/jofaichow/



IA_AI

Joined 7 years ago

"THE NMR LIFE CHOSE ME."

TC Rank

64th

of 2,023

CORR Rank

24th

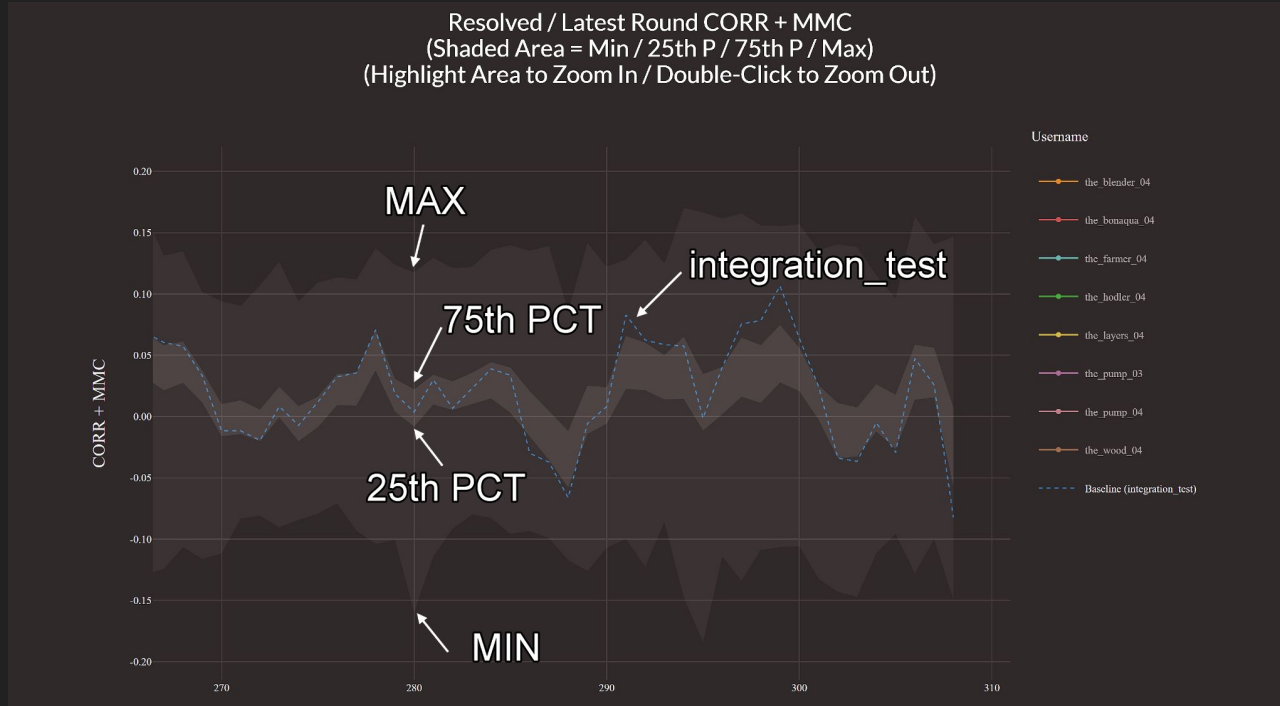
of 2,023

About this Talk

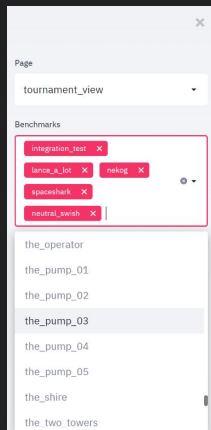
- A Brief History of My Numerati Dashboard(s)
 - Version 1 + Colab with Ceunen
 - Version 2 - Shiny Numerati
- Dashboard Use Cases
 - KPI Comparison
 - Payout Overview
 - Benchmark Models

A Brief History of My Dashboards

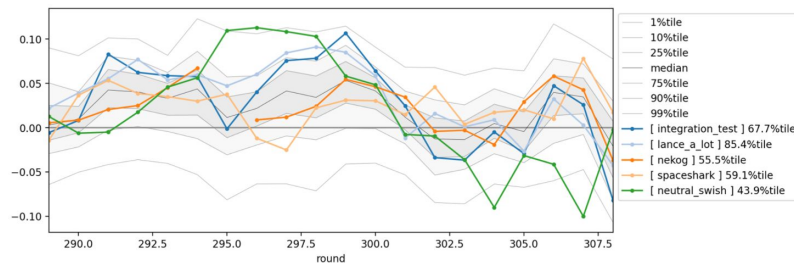
Version One (2020 - early 2022): Numerati Dashboard



First Community Colab: Numerati + BC's Streamlit



Tournament correlations/mmc



	COZIMMC
lance_a_lot	0.0357
integration_test	0.0269
spaceshark	0.0235
nekog	0.0220
neutral_swish	0.0171



Bouwe Ceunen

@BouweCeunen Follows you

Creator of [@NumeraiPayouts](#). Msc. Computer Science. Writing on Medium [medium.com/@bouwe.ceunen](#). ML/DevOps/Software Engineer.



PAYOUTS

dashboard.numeraipayouts.com

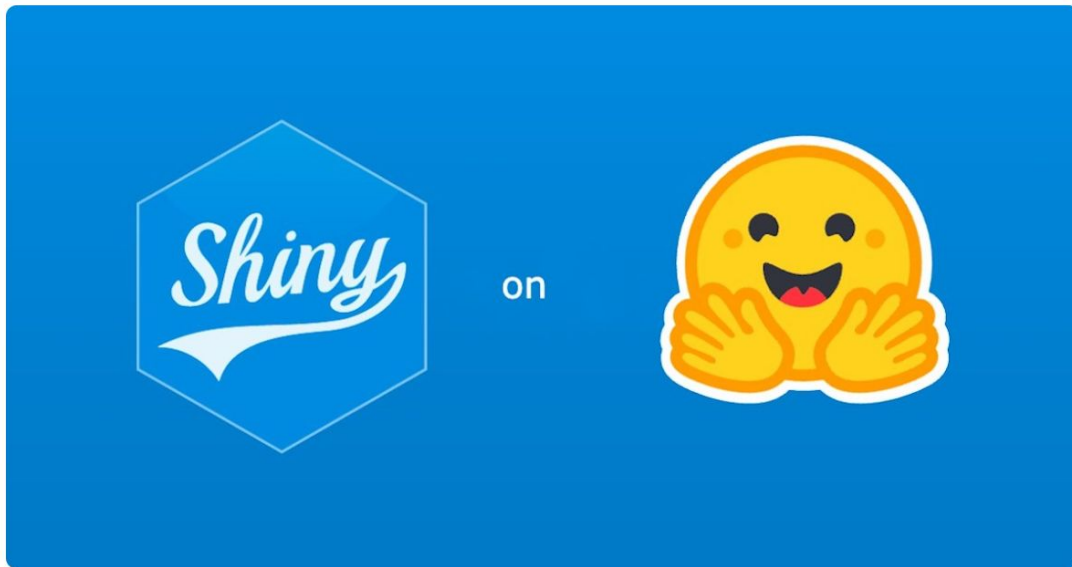
Problems with the First Version of Numerati Dashboard

- No. of models: 300 → 16000+ 
- Standalone HTML: 1MB → 30 MB (GitHub 🙋)
- Key metrics: MMC → TC, CORRv2
- ... so I stopped maintaining the project in early 2022

Shiny on Hugging Face

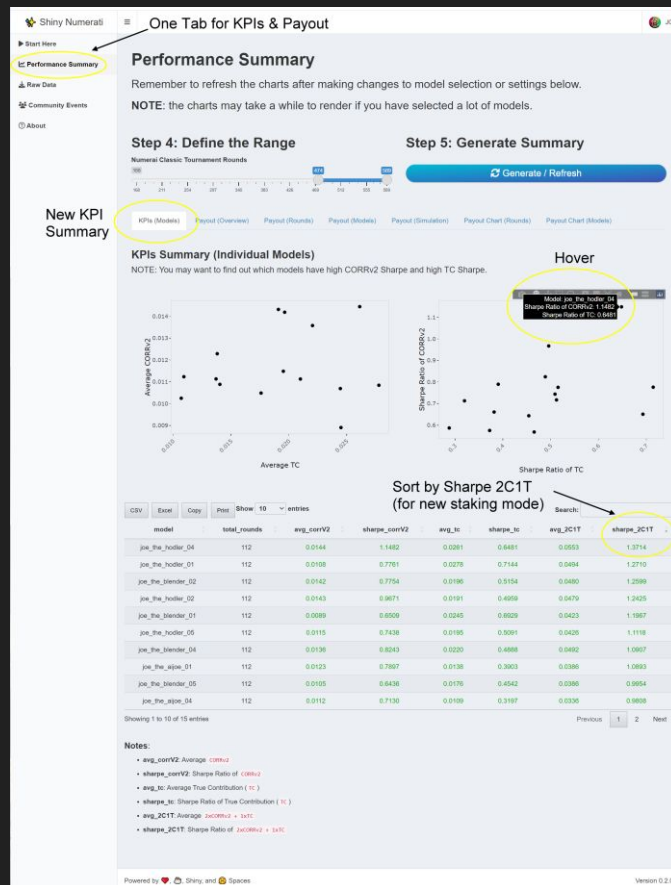
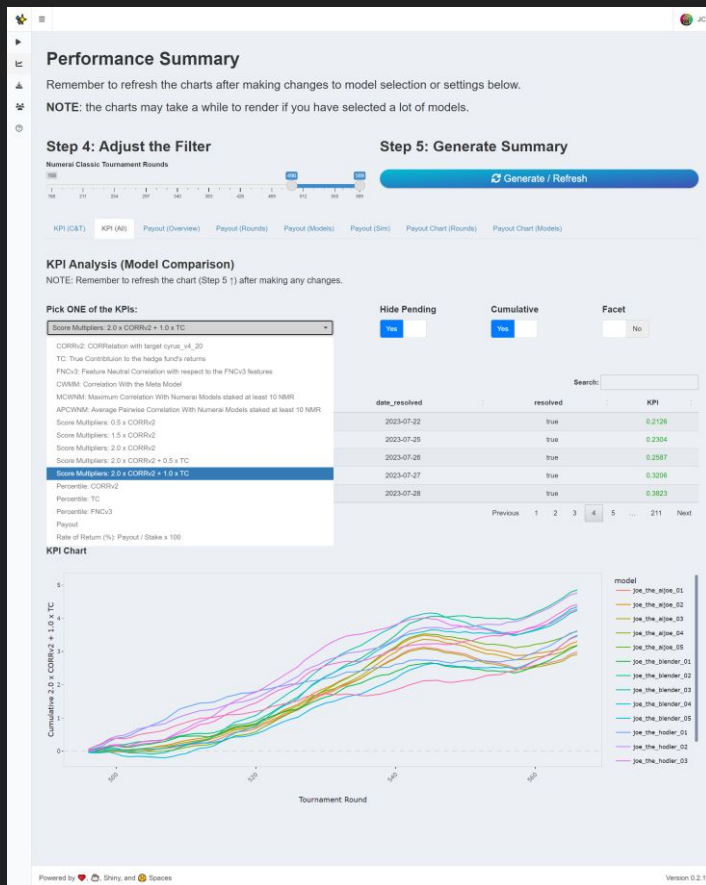
GORDON SHOTWELL

MAY 8, 2023



We're thrilled to announce that we've partnered with [Hugging Face](#) to provide templated Spaces for both the R and Python Shiny packages. These Spaces allow you to deploy an R or Python Shiny app on Hugging Face with a click of a button, and make it easy to integrate Hugging Face models into your app.

Version Two: Shiny Numerati



Latest Stake Multiplier Change

Stake multiplier updates starting round 589

■ Announcements



pschork ♥

6d

We will be updating stake multipliers starting at round 589

- Numerai users will be able to stake up to 2xCorr
- Max TC multiplier will be lowered to 1xTC for both tournaments
- Staking on TC will require 2xCorr|Fnc multiplier and will fallback to 0xTC if not set

When can users make these changes to the new multipliers?

After round 589 opens Sat Oct 7

I want to stake on TC, what do I need to do after round 589 opens?

Visit the website and ensure that you are staking on 2xCorr + 1xTC after round 589 opens.

What if I do nothing?

Your multipliers will fallback within the new ranges. If you are not staking 2xCorr, your TC multiplier will fallback to 0xTC

I'm staking 0xCorr + 3xTC now. What happens if I do nothing?

Your TC multiplier will fallback to zero at round 589 and your stake will not earn payouts or burns.

I'm staking 2xFnc + 3xTC on signals. What happens if I do nothing?

Your TC multiplier will fallback to 1xTC at round 589.

I'm staking 1xCorr + 3xTC now. What happens if I do nothing?

Your TC multiplier will fallback to 0xTC at round 589



ia_ai_Joe 06/10/2023 13:28



Using Shiny Numerati to Find Your Best Models

Live Demo →

<https://jofaichow-shiny-numerati.hf.space/>

▶ Start Here

📄 Performance Summary

📄 Raw Data

👥 Community Events

🔍 About

Shiny Numerati

Community Dashboard for the Numerai Classic Tournament

Step 1: Select Your Models

First, click this ↓

70 models selected (out of 16499)

Notes: 1) Use the search box below to find and select your models. 2) Use 'Select All' for quick selection. ✕

Select All

Deselect All

joe_the_aijoe_01



joe_the_aijoe_02



joe_the_aijoe_03



joe_the_aijoe_04



joe_the_aijoe_05



joe_the_blender_01



joe_the_blender_02



joe_the_blender_03



joe_the_blender_04



joe_the_blender_05



joe_the_bonaqua_01



joe_the_bonaqua_02



joe_the_bonaqua_03



joe_the_bonaqua_04



joe_the_bonaqua_05



Step 2: Download Data

Next, click this ↓ (it may take a while)



Download Data from Numerai

```
[49] "joe_the_layers_04" "joe_the_layers_05" "joe_the_pump_01"
[52] "joe_the_pump_02" "joe_the_pump_03" "joe_the_pump_04"
[55] "joe_the_pump_05" "joe_the_purist_01" "joe_the_purist_02"
[58] "joe_the_purist_03" "joe_the_purist_04" "joe_the_purist_05"
[61] "joe_the_sooother_01" "joe_the_sooother_02" "joe_the_sooother_03"
[64] "joe_the_sooother_04" "joe_the_sooother_05" "joe_the_wood_01"
[67] "joe_the_wood_02" "joe_the_wood_03" "joe_the_wood_04"
[70] "joe_the_wood_05"
```

Data Preview:

CSV	Excel	Copy	Print	Show 10	▼	entries										Search:																
model	↑	round	↑	date_open	↑	date_resolved	↑	resolved	↑	stake	↑	corrV2	↑	corrV2_pct	↑	fncV3	↑	fncV3_pct	↑	tc	↑	tc_pct	↑	corr_meta	↑	apcwnm	↑	mcwnm	↑	pay_ftr	↑	payout
joe_the_aijoe_01		352		2022-11-10		2022-12-13		true		10.00		0.0226		65.2		0.0371		93.1		0.0923		96.5		0.5336		0.3506		0.7855		0.0000		0.00
joe_the_aijoe_01		353		2022-11-11		2022-12-14		true		10.00		0.0445		92.4		0.0470		92.4		0.0636		90.6		0.5333		0.3530		0.7826		0.0000		0.00
joe_the_aijoe_01		354		2022-11-12		2022-12-15		true		10.00		0.0492		87.6		0.0525		90.9		0.0283		68.3		0.5476		0.3636		0.7842		0.3578		0.40
joe_the_aijoe_01		355		2022-11-15		2022-12-16		true		10.00		0.0487		86.1		0.0488		90.0		0.1015		98.2		0.5188		0.3581		0.7935		0.0000		0.00
joe_the_aijoe_01		356		2022-11-16		2022-12-17		true		10.00		0.0269		54.5		0.0270		69.9		0.0566		83.1		0.5333		0.3612		0.7898		0.0000		0.00
joe_the_aijoe_01		357		2022-11-17		2022-12-20		true		10.00		0.0345		56.5		0.0378		82.5		0.0573		58.9		0.5264		0.3522		0.7929		0.0000		0.00
joe_the_aijoe_01		358		2022-11-18		2022-12-21		true		10.00		0.0367		62.0		0.0331		71.8		0.0728		73.2		0.5339		0.3540		0.7957		0.0000		0.00
joe_the_aijoe_01		359		2022-11-19		2022-12-22		true		10.00		0.0256		60.0		0.0375		87.4		0.1102		94.9		0.5361		0.3497		0.7915		0.3676		0.64
joe_the_aijoe_01		360		2022-11-22		2022-12-23		true		10.00		0.0319		68.6		0.0462		87.2		0.0998		76.1		0.5122		0.3471		0.7926		0.0000		0.00
joe_the_aijoe_01		361		2022-11-23		2022-12-24		true		10.00		0.0305		53.9		0.0546		81.9		0.0626		47.6		0.5148		0.3494		0.7890		0.0000		0.00
Showing 121 to 130 of 19,263 entries																																
																								Previous	1	...	12	13	14	...	1,927	Next

Step 3: Performance Summary (see ←)

Note: you can also download [Raw Data] and check out our [Community Events] (see ←)

Performance Summary

Remember to refresh the charts after making changes to model selection or settings below.

NOTE: the charts may take a while to render if you have selected a lot of models.

Step 4: Adjust the Filter

Numerai Classic Tournament Rounds



Step 5: Generate Summary

[Generate / Refresh](#)

KPI (C&T)

KPI (All)

Payout (Overview)

Payout (Rounds)

Payout (Models)

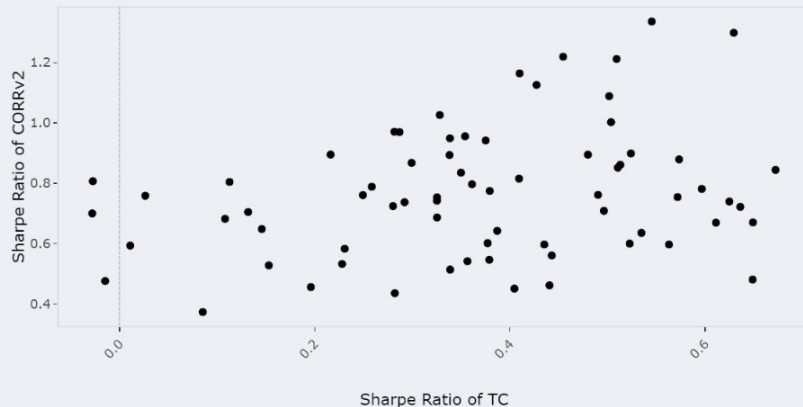
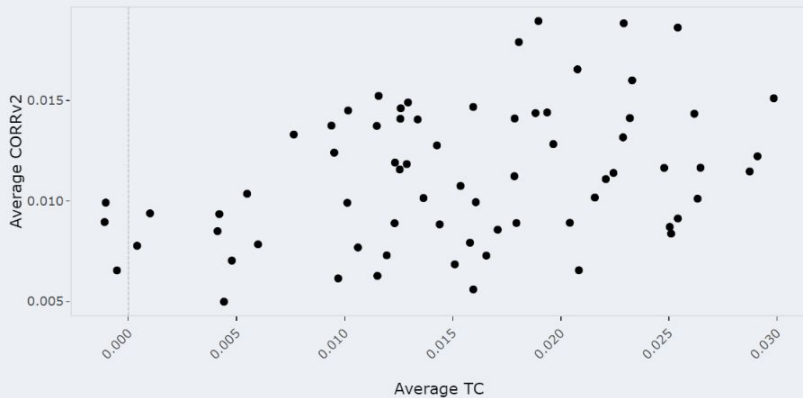
Payout (Sim)

Payout Chart (Rounds)

Payout Chart (Models)

KPI Analysis (CORRv2 and TC)

NOTE: You may want to find out which models have high CORRv2 Sharpe and high TC Sharpe.



Performance Summary

Remember to refresh the charts after making changes to model selection or settings below.

NOTE: the charts may take a while to render if you have selected a lot of models.

Step 4: Adjust the Filter

Numerai Classic Tournament Rounds



Step 5: Generate Summary



KPI (C&T)

KPI (All)

Payout (Overview)

Payout (Rounds)

Payout (Models)

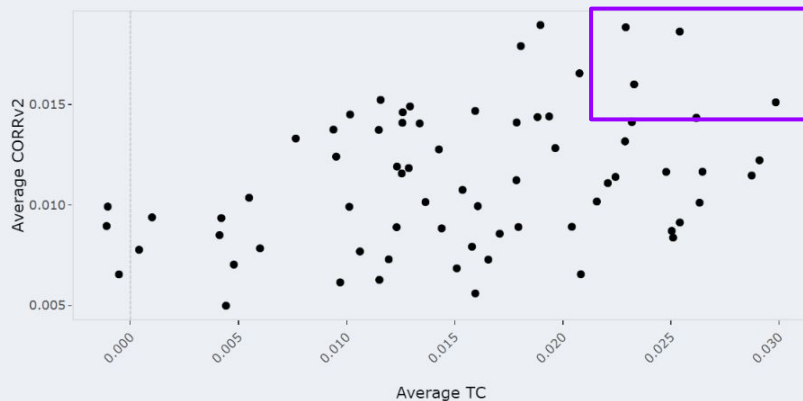
Payout (Sim)

Payout Chart (Rounds)

Payout Chart (Models)

KPI Analysis (CORRv2 and TC)

NOTE: You may want to find out which models have high CORRv2 Sharpe and high TC Sharpe.



Average TC

Sharpe Ratio of TC

CSV Excel Copy Print Show 10 entries

Search:

model	total_rounds	avg_corrV2	sharpe_corrV2	avg_tc	sharpe_tc	avg_2C1T	sharpe_2C1T
joe_the_blender_02	94	0.0190	1.2195	0.0190	0.4548	0.0569	1.3641
joe_the_ia_ai_03	94	0.0186	1.3362	0.0254	0.5456	0.0626	1.3456
joe_the_ia_ai_05	94	0.0188	1.0025	0.0229	0.5038	0.0606	1.3328
joe_the_hodler_02	94	0.0165	1.2122	0.0208	0.5095	0.0541	1.3269
joe_the_bonaqua_03	94	0.0144	0.8611	0.0188	0.5133	0.0476	1.2973
joe_the_coin_03	94	0.0151	1.2989	0.0298	0.6296	0.0604	1.2749
joe_the_bonaqua_02	94	0.0141	0.8946	0.0179	0.4802	0.0461	1.2385
joe_the_farmer_03	94	0.0122	0.8443	0.0291	0.6725	0.0535	1.2377
joe_the_hodler_05	94	0.0111	0.6693	0.0221	0.6113	0.0444	1.2304
joe_the_ia_ai_01	94	0.0179	1.1643	0.0181	0.4101	0.0539	1.2236

Showing 1 to 10 of 70 entries

Previous

1

2

3

4

5

6

7

Next

Notes:

- **avg_corrV2**: Average CORRV2
- **sharpe_corrV2**: Sharpe Ratio of CORRV2
- **avg_tc**: Average True Contribution (TC)
- **sharpe_tc**: Sharpe Ratio of True Contribution (TC)
- **avg_2C1T**: Average $2 \times \text{CORRV2} + 1 \times \text{TC}$
- **sharpe_2C1T**: Sharpe Ratio of $2 \times \text{CORRV2} + 1 \times \text{TC}$

Step 4: Adjust the Filter

Numerai Classic Tournament Rounds



Step 5: Generate Summary

 Generate / Refresh

KPI (C&T)

KPI (All)

Payout (Overview)

Payout (Rounds)

Payout (Models)

Payout (Sim)

Payout Chart (Rounds)

Payout Chart (Models)

KPI Analysis (Model Comparison)

NOTE: Remember to refresh the chart (Step 5 ↑) after making any changes.

Pick ONE of the KPIs:

CORRv2: CORRelation with target cyrus_v4_20

CORRv2: CORRelation with target cyrus_v4_20

TC: True Contribution to the hedge fund's returns

FNCv3: Feature Neutral Correlation with respect to the FNCv3 features

CWMM: Correlation With the Meta Model

MCWNM: Maximum Correlation With Numerai Models staked at least 10 NMR

APCWNM: Average Pairwise Correlation With Numerai Models staked at least 10 NMR

Score Multipliers: 0.5 x CORRv2

Score Multipliers: 1.5 x CORRv2

Score Multipliers: 2.0 x CORRv2

Score Multipliers: 2.0 x CORRv2 + 0.5 x TC

Score Multipliers: 2.0 x CORRv2 + 1.0 x TC

Percentile: CORRv2

Percentile: TC

Percentile: FNCv3

Payout

Rate of Return (%): Payout / Stake x 100

Cumulative

Yes

Hide Pending

No

Facet

No

Search:

date_resolved

resolved

KPI

2023-07-01

true

0.0024

2023-07-04

true

0.0045

2023-07-05

true

0.0091

2023-07-06

true

0.0151

2023-07-07

true

0.0292

Previous

1

2

3

4

5

...

1,316

Next

KPI Chart (Remember to Click the "Refresh" Button !!!)

2.0

model

KPI Data

CSV Excel Copy Print Show 5 entries

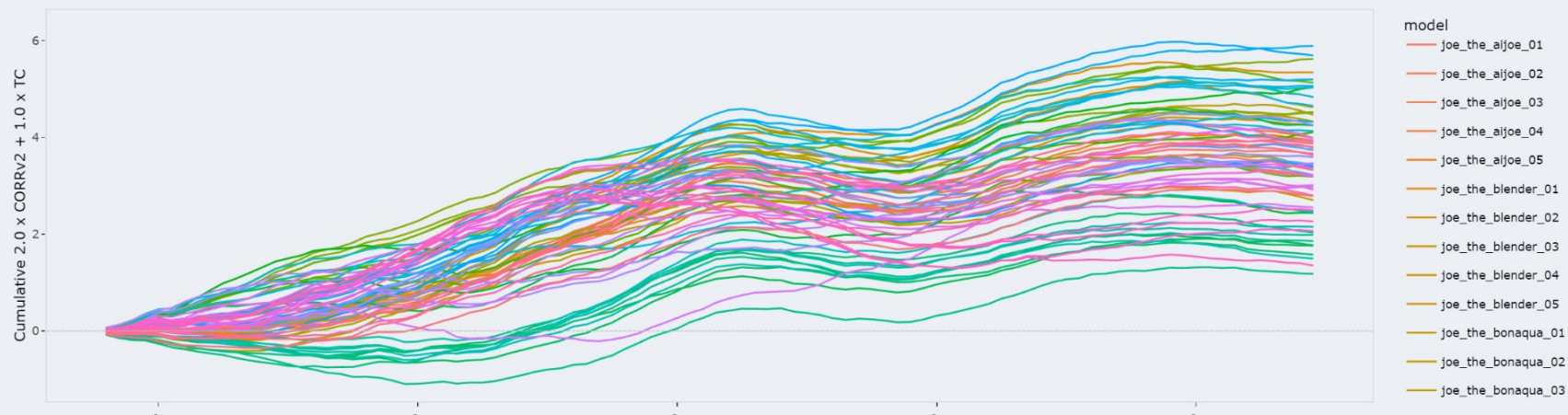
Search:

model	round	date_resolved	resolved	KPI
joe_the_aijoe_01	511	2023-07-22	true	0.1438
joe_the_aijoe_01	512	2023-07-25	true	0.1847
joe_the_aijoe_01	513	2023-07-26	true	0.2134
joe_the_aijoe_01	514	2023-07-27	true	0.2816
joe_the_aijoe_01	515	2023-07-28	true	0.3246

Showing 16 to 20 of 6,560 entries

Previous 1 2 3 4 5 ... 1,312 Next

KPI Chart (Remember to Click the 'Refresh' Button ↑↑↑)



Pick ONE of the KPIs:

Score Multipliers: 2.0 x CORRV2 + 1.0 x TC

Cumulative

Yes

Hide Pending

No

Facet

Yes

KPI Data

CSV Excel Copy Print Show 5 entries

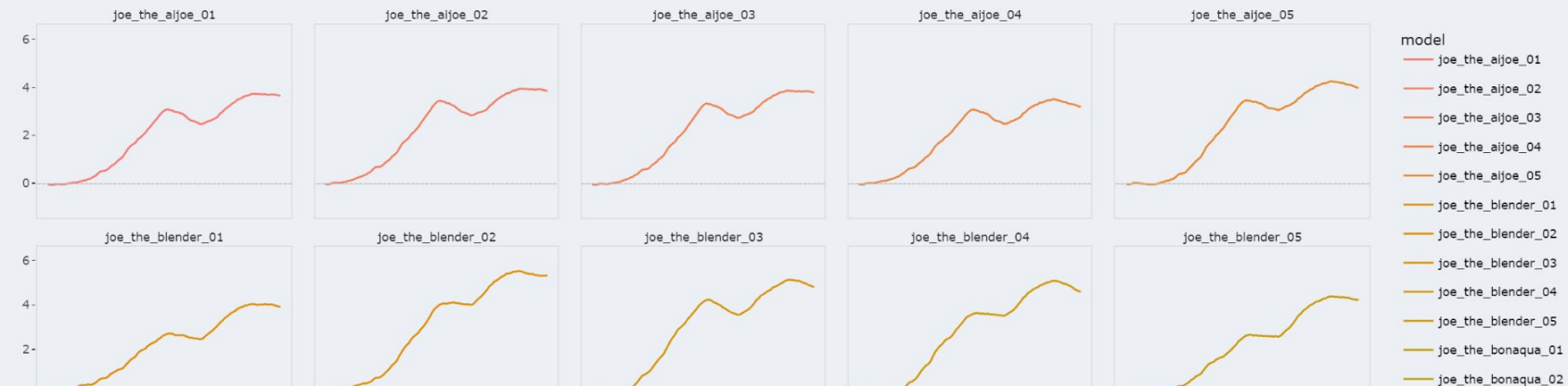
Search:

model	round	date_resolved	resolved	KPI
joe_the_aijoe_01	511	2023-07-22	true	0.1438
joe_the_aijoe_01	512	2023-07-25	true	0.1847
joe_the_aijoe_01	513	2023-07-26	true	0.2134
joe_the_aijoe_01	514	2023-07-27	true	0.2816
joe_the_aijoe_01	515	2023-07-28	true	0.3246

Showing 16 to 20 of 6,560 entries

Previous 1 2 3 4 5 ... 1,312 Next

KPI Chart (Remember to Click the 'Refresh' Button ↑↑↑)



75

Staked Rounds (Resolved)

2375.28 NMR

Total Payout (Resolved)

31.67 NMR

Avg. Round Payout (Resolved)

0.70 %

Avg. Round ROR (Resolved)

19

Staked Rounds (Pending)

111.71 NMR

Total Payout (Pending)

5.88 NMR

Avg. Round Payout (Pending)

0.10 %

Avg. Round ROR (Pending)

94

Staked Rounds (All)

2486.99 NMR

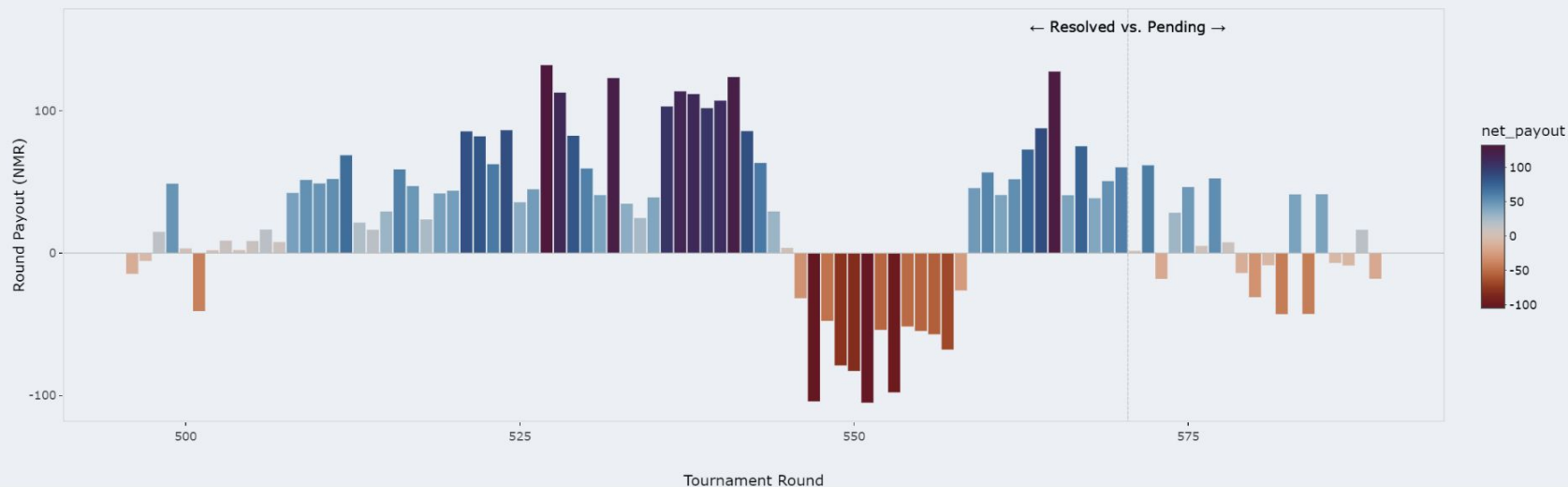
Total Payout (All)

26.46 NMR

Avg. Round Payout (All)

0.58 %

Avg. Round ROR (All)



Payout Simulation (Overall)

sum_pay_1C0T	sum_pay_2C0T	sum_pay_2C1T	sum_pay_1C3T	shp_pay_1C0T	shp_pay_2C0T	shp_pay_2C1T	shp_pay_1C3T
556.94	1,113.87	1,760.72	2,497.49	0.69	0.69	0.58	0.39

Payout Simulation (Individual Models)

CSVExcelCopyPrintShow100▼entriesSearch:

model	rounds	sum_pay_1C0T	sum_pay_2C0T	sum_pay_2C1T	sum_pay_1C3T	shp_pay_1C0T	shp_pay_2C0T	shp_pay_2C1T	shp_pay_1C3T
joe_the_aijoe_01	93	9.95	19.90	26.57	29.97	0.89	0.89	0.62	0.34
joe_the_aijoe_02	93	9.42	18.85	27.48	35.33	0.87	0.87	0.63	0.39
joe_the_aijoe_03	93	9.59	19.17	27.36	34.15	0.87	0.87	0.63	0.39
joe_the_aijoe_04	93	8.83	17.66	21.97	21.77	0.79	0.79	0.52	0.25
joe_the_aijoe_05	93	10.14	20.27	28.52	34.89	0.82	0.82	0.63	0.39
joe_the_blender_01	94	8.15	16.30	29.61	48.06	0.75	0.75	0.76	0.57
joe_the_blender_02	94	13.42	26.84	39.16	50.38	1.14	1.14	0.84	0.51
joe_the_blender_03	94	9.25	18.51	33.78	55.08	0.69	0.69	0.63	0.50
joe_the_blender_04	94	9.09	18.18	33.28	54.38	0.62	0.62	0.66	0.54
joe_the_blender_05	94	9.78	19.55	33.10	50.40	0.79	0.79	0.78	0.57
joe_the_bonaqua_01	94	4.90	9.80	18.33	30.48	0.45	0.45	0.49	0.41
joe_the_bonaqua_02	94	9.71	19.42	30.35	42.50	0.86	0.86	0.73	0.51
joe_the_bonaqua_03	94	9.73	19.46	31.70	46.46	0.78	0.78	0.72	0.56
joe_the_bonaqua_04	94	6.64	13.27	23.03	35.93	0.50	0.50	0.48	0.39
joe_the_bonaqua_05	94	6.03	12.05	22.69	37.93	0.51	0.51	0.53	0.44
joe_the_coin_01	93	9.88	19.75	30.20	41.22	1.04	1.04	0.76	0.43
joe_the_coin_02	93	8.05	16.10	29.68	48.78	0.81	0.81	0.74	0.51

NOTE: the charts may take a while to render if you have selected a lot of models.

Step 4: Adjust the Filter

Numerai Classic Tournament Rounds



Step 5: Generate Summary

 Generate / Refresh

KPI (C&T)

KPI (All)

Payout (Overview)

Payout (Rounds)

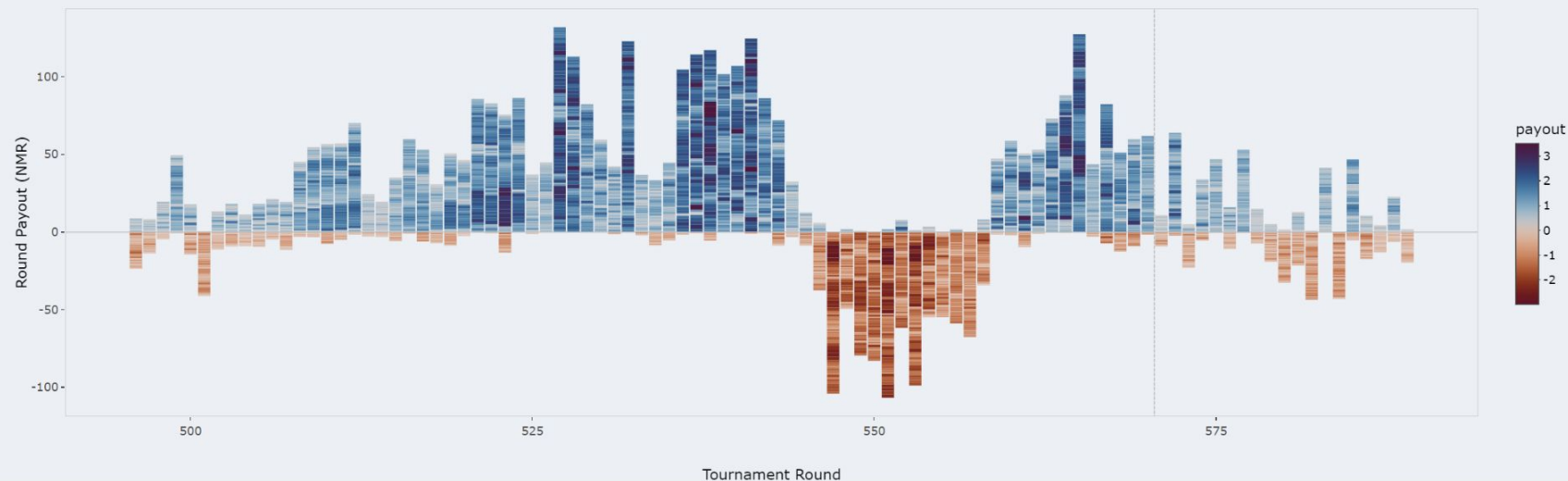
Payout (Models)

Payout (Sim)

Payout Chart (Rounds)

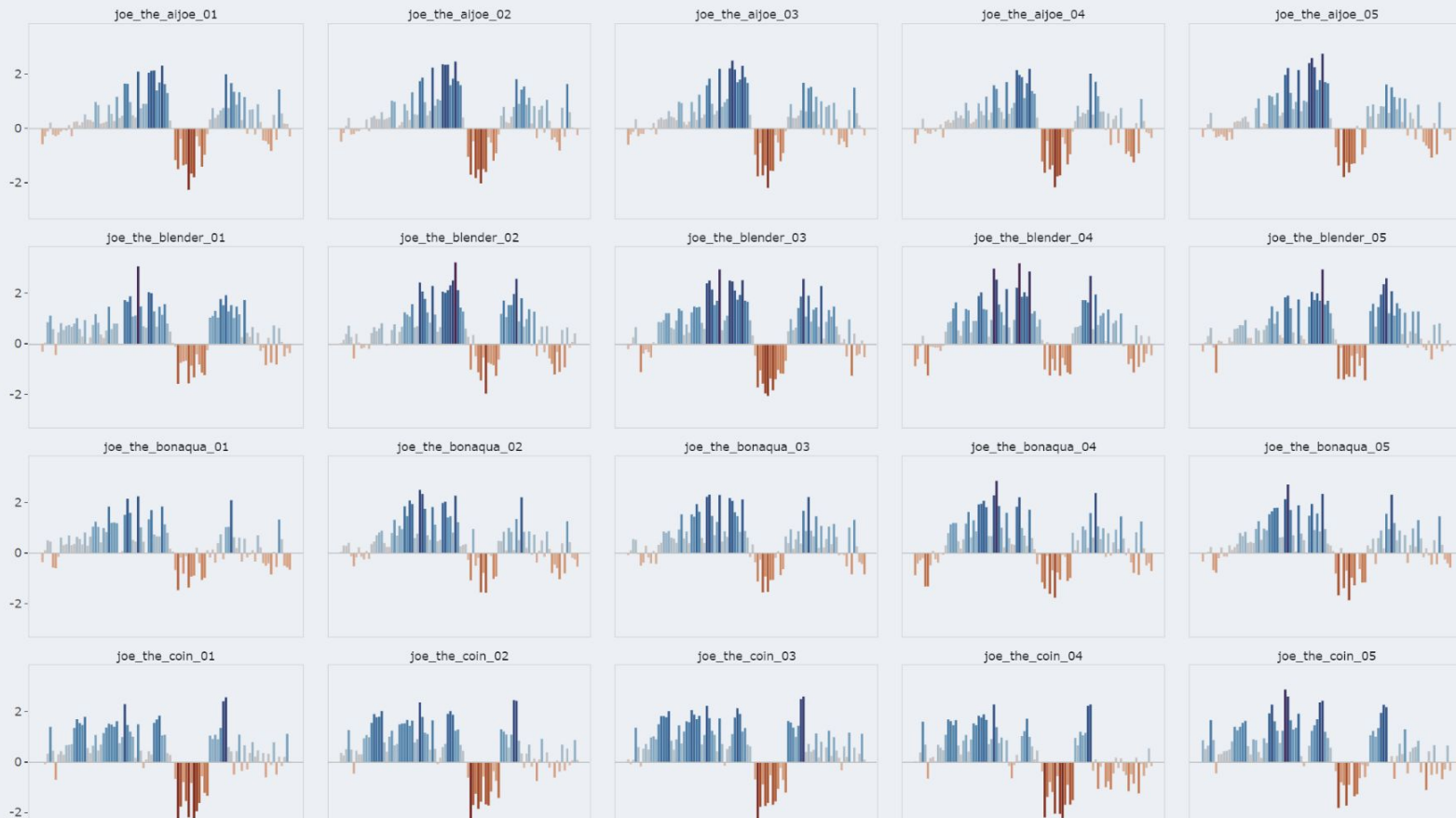
Payout Chart (Models)

Payout Summary Chart (All Models - Stacked)





Payout Summary Chart (Individual Models)



payout



@ia_ai_Joe @MikeP which existing benchmark models are your



MikeP 10/10/2023 13:15

the most relevant are

https://numer.ai/lg_lgbm_v42_cyrus20

https://numer.ai/sunshine_example

https://numer.ai/example_preds_v4

anddddddd

tbh v2 example preds are still good and interesting

https://numer.ai/example_preds_v2

model	round	date_resolved	resolved	KPI
example_preds_v2	474	2023-06-01	true	-0.0141
example_preds_v2	475	2023-06-02	true	-0.0290
example_preds_v2	476	2023-06-03	true	-0.0192
example_preds_v2	477	2023-06-06	true	-0.0225
example_preds_v2	478	2023-06-07	true	-0.0224

Showing 1 to 5 of 464 entries

Previous

1

2

3

4

5

...

93

Next

KPI Chart (Remember to Click the 'Refresh' Button ↑↑↑)



97

Staked Rounds (Resolved)

-0.82 NMR

Total Payout (Resolved)

-0.01 NMR

Avg. Round Payout (Resolved)

-0.05 %

Avg. Round ROR (Resolved)

19

Staked Rounds (Pending)

-0.11 NMR

Total Payout (Pending)

-0.01 NMR

Avg. Round Payout (Pending)

-0.04 %

Avg. Round ROR (Pending)

116

Staked Rounds (All)

-0.93 NMR

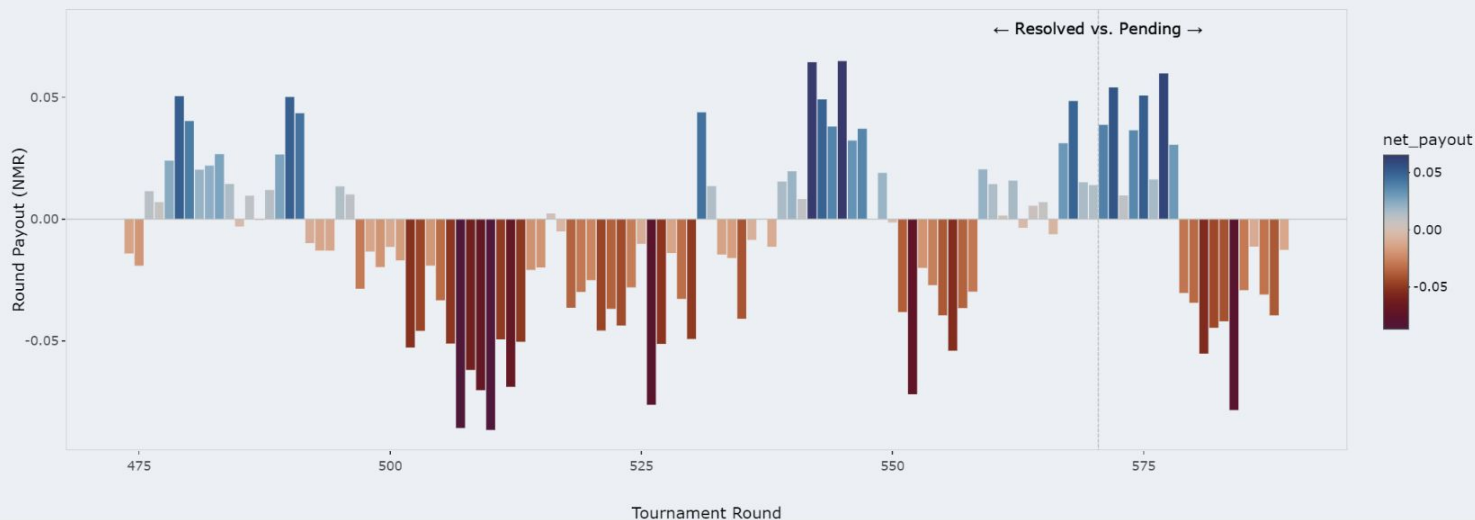
Total Payout (All)

-0.01 NMR

Avg. Round Payout (All)

-0.05 %

Avg. Round ROR (All)

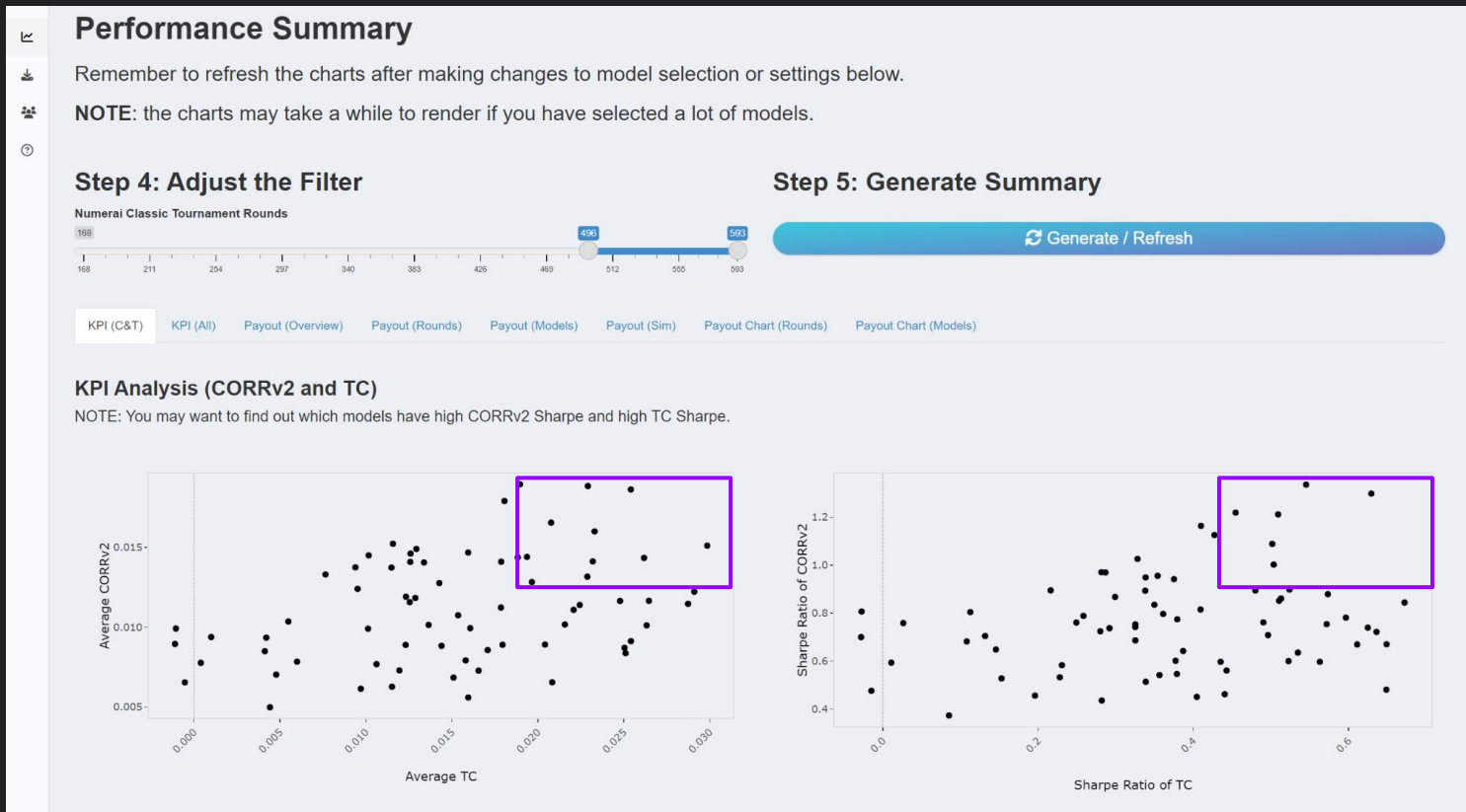


Try it out →

<https://jofaichow-shiny-numerati.hf.space/>

Feedback / Suggestion:
Numerai Discord: ia_ai_Joe

Conclusions - Find Models with High Corr & TC



Conclusions - R.I.P. 3xTC



ia_ai_Joe 12/10/2023 14:06

The notorious 3xTC were once roaming the earth like no other degen. Now they are living in our memories 🥹🥹🥹
(Royal Ontario Museum) (edited)



ia_ai_Joe 12/10/2023 18:35

My 1xCORR+3xTC payout vs. 2xCORR+1xTC payout (edited)



Kenny Lay (Parody) ✓
@EnronChairman

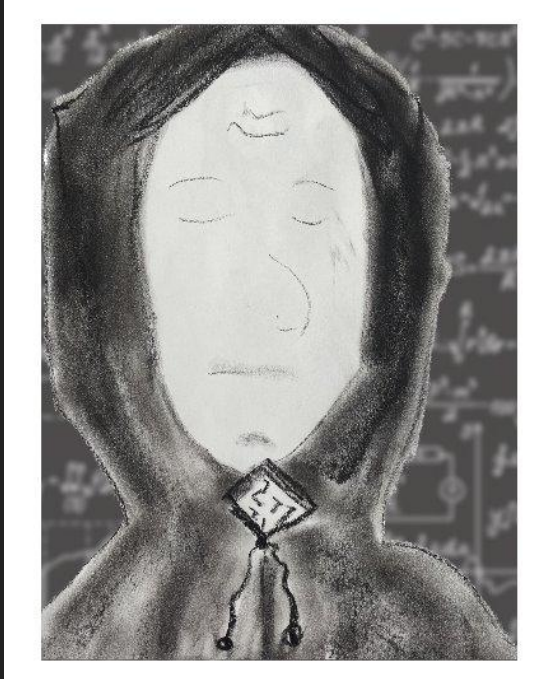
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Your investment strategy vs. investment return



Any Questions?



Contact

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